COMPARATIVE ANALYSIS OF CHEMICAL SPACE COVERAGE FOR GLUCOCORTICOID RECEPTOR LIGANDS

Datasets: Reference (ChEMBL2034), DrugEx_GT_epsilon_0.6, GB_GA_mut_r_0.5

Introduction

Chemical space visualization provides insight into how generative models explore molecular diversity relative to known ligands. Three datasets were analyzed: the ChEMBL2034 reference set, molecules generated by DrugEx_GT_epsilon_0.6, and those produced by GB_GA_mut_r_0.5. Two complementary representations were used: Morgan fingerprints (radius 2, 2048 bits) for structural similarity. Physicochemical descriptors (MolWt, LogP, TPSA, HBD, HBA, RotB) for property-based similarity. Dimensionality reduction was performed using PCA and t-SNE to visualize the datasets in two dimensions.

Discussion & Results

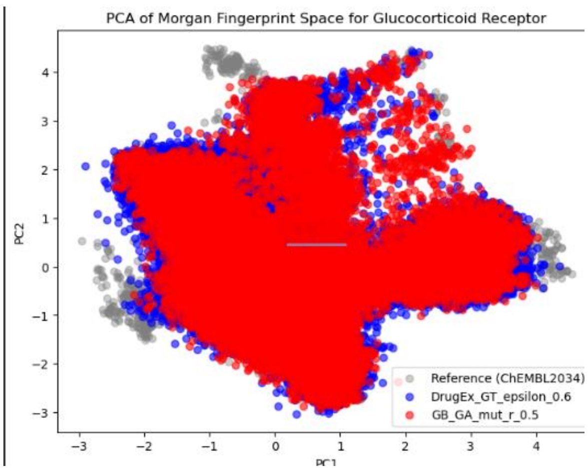


Figure 1a. Morgan Fingerprint PCA

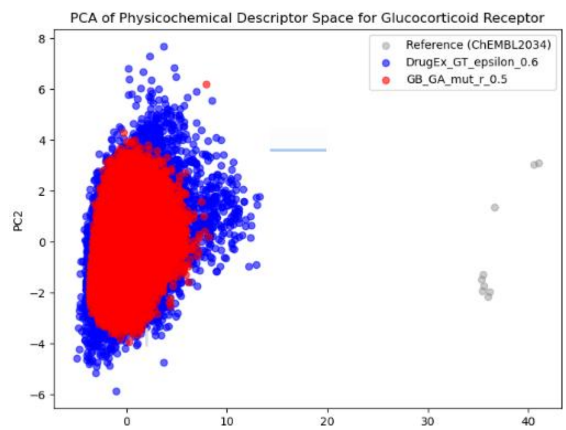


Figure 1b. Descriptors PCA

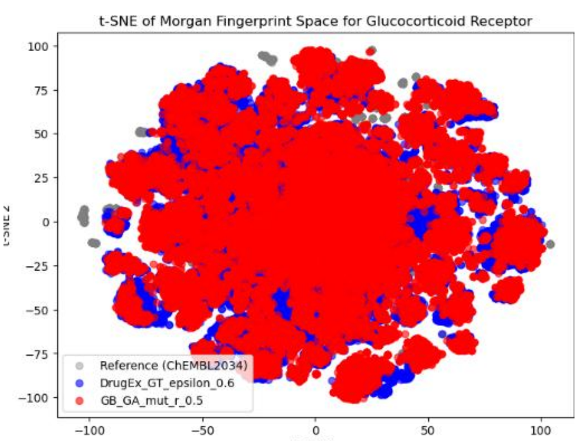


Figure 2a. Morgan Fingerprint t-SNE

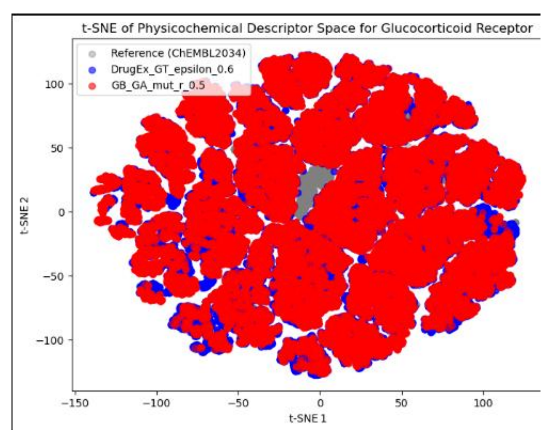


Figure 2b. Descriptors t-SNE

1. Which datasets overlap strongly, and which are clearly separated?

Across both PCA and t-SNE plots, the GB_GA_mut_r_0.5 (red) and DrugEx_GT_epsilon_0.6 (blue) datasets overlap extensively with each other and with the reference (gray) set. The overlap is strongest in the fingerprint-based plots, where all three datasets occupy a shared central region. A clearer separation is evident in the descriptor-based t-SNE, where DrugEx molecules extend further outward, indicating the exploration of new physicochemical regions.

2. Do the generated sets stay close to reference chemistry, or do they explore a different region?

Both generated sets remain anchored near the reference chemistry, confirming that they learned glucocorticoid-relevant features. However, DrugEx_GT_epsilon_0.6 explores a broader structural and physicochemical space, which is visible as peripheral clusters and wider dispersion, while GB_GA_mut_r_0.5 stays tightly clustered around the reference, suggesting conservative optimization rather than exploration.

3. **Which method, PCA or t-SNE, makes the cluster structure easier to see?**

The t-SNE plots make cluster structure and density differences easier to interpret. t-SNE emphasizes local relationships, revealing compact clusters and peripheral expansions that PCA compresses into overlapping ellipses. PCA is useful for global variance trends, but t-SNE provides clearer visual separation between generated and reference regions.

4. Do you observe outliers, and if so, what might explain them?

Yes. A few gray reference points appear as outliers in the descriptor-based PCA, far from the main cluster. These likely correspond to atypical glucocorticoid ligands with extreme molecular weights, polar surface areas, or unusual hydrogen-bonding profiles.

5. How does the interpretation change when using descriptors instead of fingerprints?

Using descriptors shifts the focus from structural similarity to physicochemical property similarity. Descriptor-based plots show tighter clustering and less dispersion, indicating that most molecules (even structurally diverse ones) share comparable drug-like properties. Fingerprint-based plots, in contrast, reveal broader structural diversity and scaffold exploration. Thus, fingerprints highlight novel structural motifs, while descriptors emphasize property conservation across datasets.

Scoring Analysis of GR Ligand Generators

The custom score was constructed by normalizing each component relative to the reference dataset using MinMaxScaler. Predicted pIC_{50} values were scaled to the 0–1 range, synthetic accessibility scores were normalized and inverted so that synthetically easier molecules received higher values, QED was used directly since it already lies between 0 and 1, and the steroid-likeness penalty was inverted to reward non-steroidal scaffolds. These normalized components were combined using a weighted sum: 0.25 for QED, 0.25 for SA, 0.30 for pIC_{50}, and 0.20 for the penalty. This weighting scheme emphasizes predicted biological activity while maintaining drug-likeness and synthetic feasibility.

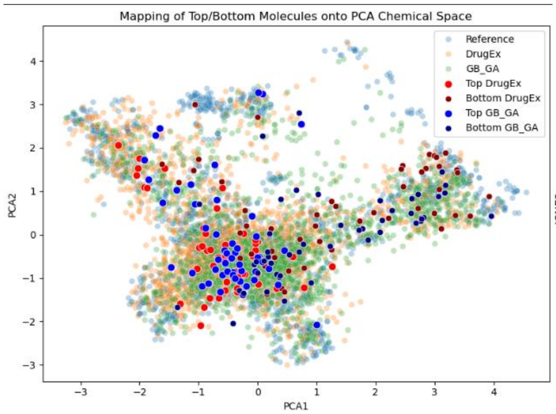
Across the individual components, GB_GA produced the most drug-like molecules (mean QED = 0.53), followed by the reference set (0.50) and DrugEx (0.42). Synthetic accessibility followed a similar trend: the reference set was easiest to synthesize (mean SA = 3.47), GB_GA was moderately accessible (3.88), and

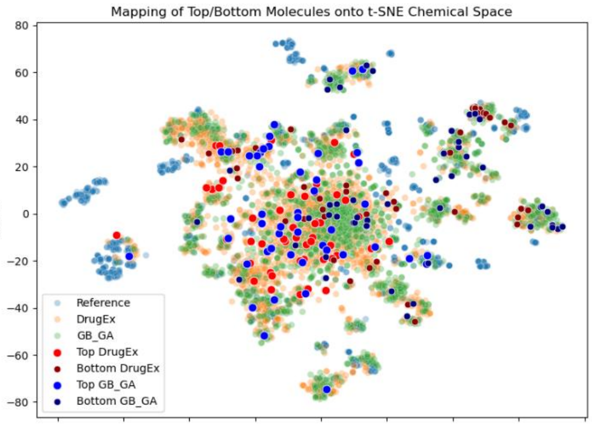
DrugEx molecules were slightly more complex (4.00). Both generative models produced less steroid-like molecules than the reference set, with penalties around 0.20 compared to 0.24 for ChEMBL2034. Predicted GR activity was highest for the reference ligands (mean $pIC_{50} = 6.97$), followed by DrugEx (6.75) and GB_GA (6.55). When these components were combined into the composite score, GB_GA achieved the highest mean value (0.78), followed by DrugEx (0.75) and the reference set (0.74), indicating that GB_GA generated the most balanced molecules overall.

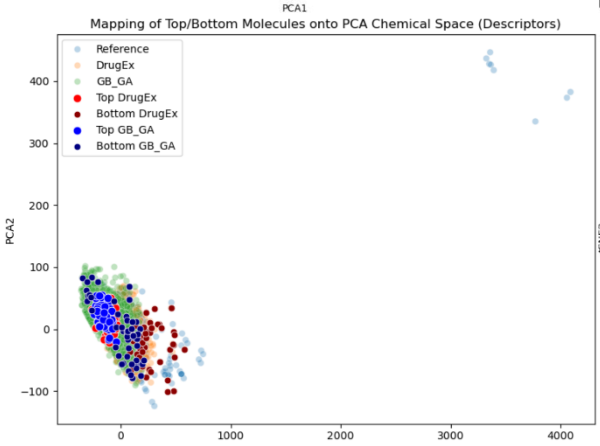
Chemical-space visualizations supported these trends. In descriptor-based PCA and t-SNE maps, reference molecules formed a dense central cluster. DrugEx molecules tended to occupy regions close to this cluster, indicating conservative optimization around known GR ligand chemistry. GB_GA molecules extended into peripheral zones associated with lower SA and lower steroid-likeness, demonstrating exploration of novel, non-steroidal chemotypes. Bottom-ranked molecules from both generators appeared at the edges of chemical space, corresponding to extreme descriptor values such as high molecular weight, high polarity, or excessive flexibility.

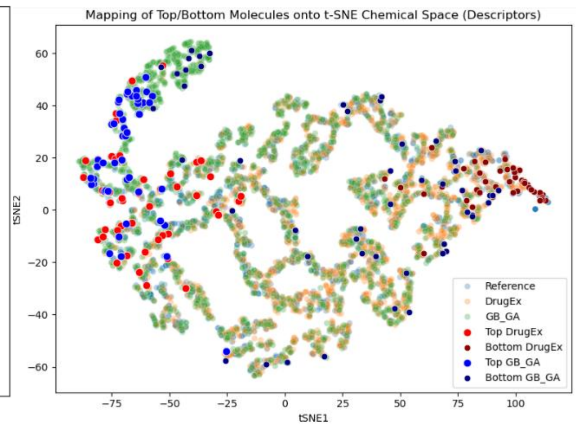
Threshold analysis was performed using literature-based criteria (QED ≥ 0.4, SA ≤ 4, penalty ≤ 0.3, and predicted $pIC_{50} \geq 6$). Code-based evaluation of the top-50 molecules from each generator showed that both DrugEx and GB_GA produced uniformly drug-like and synthetically feasible compounds, with no violations of the QED or SA thresholds. However, DrugEx had five molecules exceeding the steroid-likeness penalty and thirteen molecules with predicted pIC_{50} below 6, indicating weaker GR activity among some top-ranked candidates. In contrast, GB_GA had eight molecules above the penalty threshold but only a single molecule with predicted pIC_{50} below 6, demonstrating that GB_GA's top-ranked molecules more consistently satisfy potency criteria while remaining drug-like and synthetically accessible.

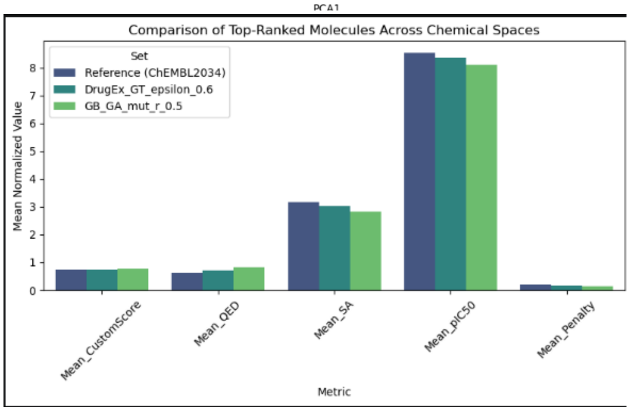
Overall, integrating the scoring analysis with chemical-space visualization provides a clear picture of generative model behavior. DrugEx tends to produce potent, reference-like GR scaffolds, while GB_GA generates more novel, non-steroidal, and synthetically accessible molecules with excellent drug-likeness. The custom score effectively captures the balance between potency, drug-likeness, synthetic feasibility, and structural novelty, and the chemical-space mapping confirms that top-ranked molecules occupy desirable regions. Taken together, GB_GA mut r 0.5 demonstrates the strongest balance of novelty and quality, whereas DrugEx GT ε 0.6 excels in producing molecules closely aligned with known GR ligand chemistry.


Mapping of Top/Bottom Molecules onto PCA Chemical Space
Reference
DrugEx
GB_GA
Top DrugEx
Bottom DrugEx
Top GB_GA
Bottom GB_GA
PCA1
PCA2


Mapping of Top/Bottom Molecules onto t-SNE Chemical Space
Reference
DrugEx
GB_GA
Top DrugEx
Bottom DrugEx
Top GB_GA
Bottom GB_GA
tSNE2


Mapping of Top/Bottom Molecules onto PCA Chemical Space (Descriptors)
Reference
DrugEx
GB_GA
Top DrugEx
Bottom DrugEx
Top GB_GA
Bottom GB_GA
PCA2
PCA1


Mapping of Top/Bottom Molecules onto t-SNE Chemical Space (Descriptors)
Reference
DrugEx
GB_GA
Top DrugEx
Bottom DrugEx
Top GB_GA
Bottom GB_GA
tSNE1
tSNE2


Comparison of Top-Ranked Molecules Across Chemical Spaces
Set
Reference (ChEMBL2034)
DrugEx_GT_epsilon_0.6
GB_GA_mut_r_0.5
Mean Normalized Value
Mean_CustomScore
Mean_QED
Mean_SA
Mean_pIC50
Mean_Penalty
Metric